Automated Business Workflow using Deluge Scripting

Application Testing and Debugging — Energy Consumption Workflows in the Smart Campus Sustainability & Carbon Tracker

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1. Aim

To design and implement an automated business-process workflow within the **Smart Campus Sustainability & Carbon Tracker** application in Zoho Creator using Deluge scripting, and to conduct application testing and debugging to verify real-time execution on the **Energy_Consumption** form: automatic carbon estimation, a high-consumption alert and an automatic energy status.

Objectives

- To configure form-level workflows that run automatically when an energy record is created or edited and validated.
- To write custom backend logic in Deluge that reads the Units_Consumed_kWh field, calculates Estimated CO_2 (units × 0.82), sets the Energy Status and blocks unusually high readings.
- To perform end-to-end debugging and live testing to ensure data accuracy and workflow reliability.

2. Procedure

Step 1: Initiating the Workflow Configuration

1. Open the Zoho Creator application in Edit mode.
2. Select Workflow from the top navigation.
3. Click Create Workflow to open the automation setup window.

Step 2: Defining the Trigger Parameters

1. Under Choose form, select the target form: Energy_Consumption.
2. For the Record Event, select Created or Edited so the automation also recalculates values when a reading is corrected.
3. For the Form Event, select On validate so the calculated values are stored together with the record in a single submission.
4. Name the workflow Calculate_Estimated_CO2 and click Create Workflow. Repeat the same trigger for High_Energy_Consumption_Alert and Energy_Status.

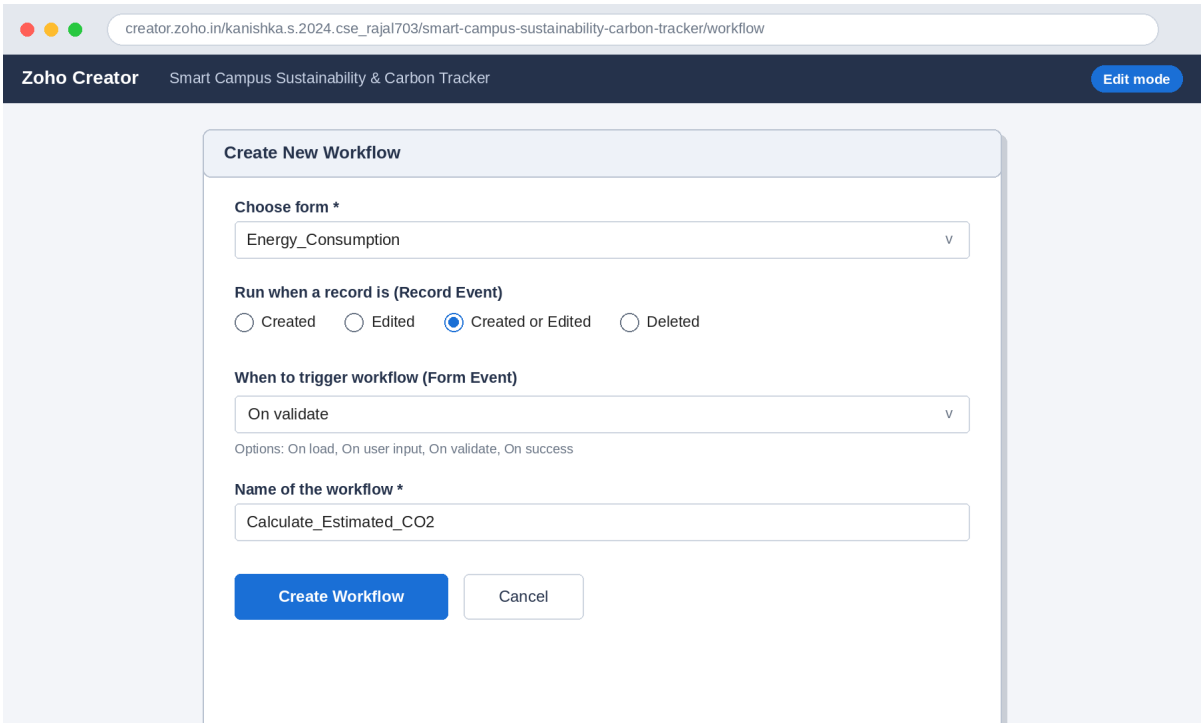


Illustration of the workflow setup window: form selected, Record Event set to Created or Edited, Form Event set to On validate, and the workflow named, ready to be created (illustration).

Explanation of the Workflow Setup Window

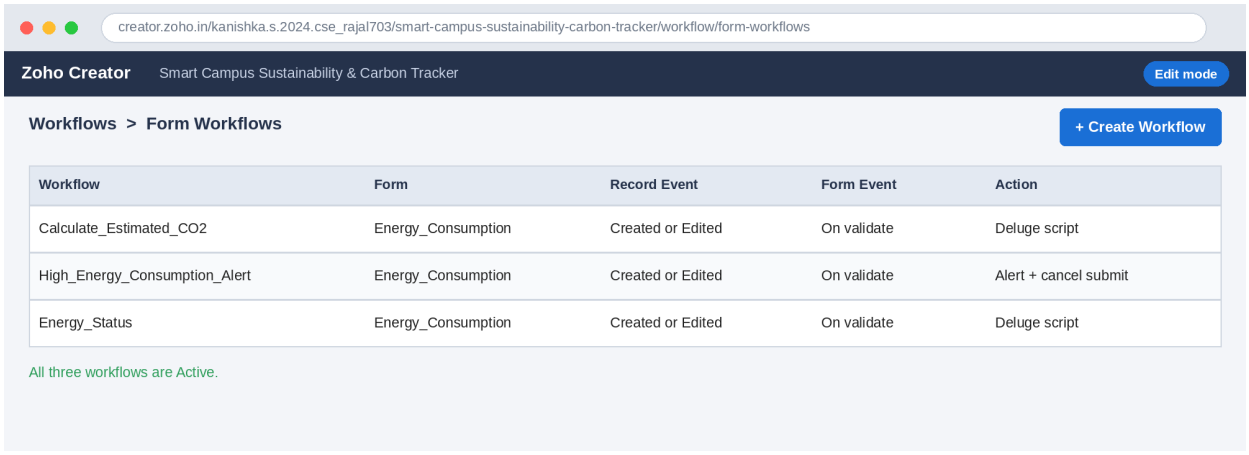
Setting	Value Selected	Purpose
Choose form	Energy_Consumption	Links the workflow to the form where electricity readings are entered, so only this form's records activate it.
Record Event	Created or Edited	Runs on a new record and on every correction, so the carbon value and status never go out of date.
Form Event	On validate	Runs just before the record is saved, so the calculated fields are saved with the same submission and a bad value can be rejected.
Workflow Name	Calculate_Estimated_CO2	A clear name that identifies the purpose of the workflow.

Why the Form Event is set to "On validate"

- The script runs before the record is stored, so Estimated CO_2 and Energy Status are saved with the record without a second update.
- The same event allows *cancel submit*, which is needed to stop readings above 1000 kWh.
- If the user edits the reading, the calculated fields are refreshed on the next save.

Observations

- After clicking Create Workflow, Zoho Creator opens the workflow canvas for the chosen form and event.
- The trigger is now active and waiting for an action to be attached.
- No errors appeared while creating the three workflows. The next step is adding the Deluge script action.



Form Workflows list showing the three Energy_Consumption workflows (illustration).

Step 3: Setting up the Action Interface

1. Inside the newly created workflow block, click Add New Action.
2. Select Deluge script from the available action types (Notification, Data access, Integrations, Deluge script) to launch the Deluge editor.

Step 4: Developing the Deluge Script

The scripts below were entered in the three workflows. They use the field link name `Units_Consumed_kWh`, and write to `Estimated_CO` and `Energy_Status`.

```
// DS-02: carbon estimate = units consumed x emission factor (0.82)
if(input.Units_Consumed_kWh > 0)
{
    input.Estimated_CO = input.Units_Consumed_kWh * 0.82;
}
else
{
    input.Estimated_CO = 0;
}
```

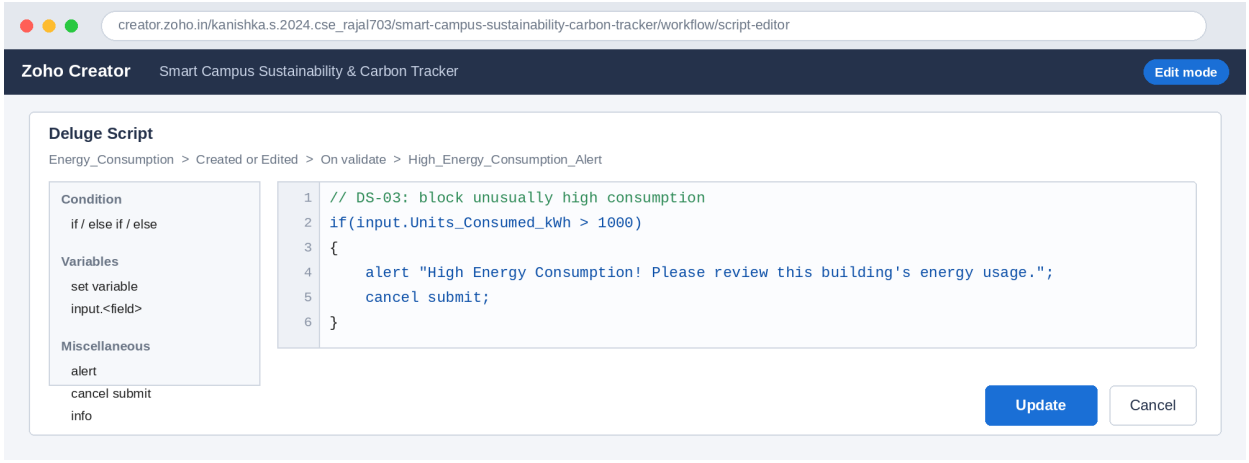
Code 6.1 — Calculate_Estimated_CO2 (DS-02)



Deluge editor for Calculate_Estimated_CO2 (illustration).

```
// DS-03: block unusually high consumption
if(input.Units_Consumed_kWh > 1000)
{
    alert "High Energy Consumption! Please review this building's energy usage.";
    cancel submit;
}
```

Code 6.2 — High_Energy_Consumption_Alert (DS-03)



Deluge editor for High_Energy_Consumption_Alert (illustration).

```
// DS-04: set the status label from the units consumed
if(input.Units_Consumed_kWh > 1000)
{
    input.Energy_Status = "High Consumption";
}
else
{
    input.Energy_Status = "Normal";
}
```

Code 6.3 — Energy_Status (DS-04)



Deluge editor for Energy_Status (illustration).

1. Write the backend code inside the editor using Zoho's Deluge syntax. Each script tests the value of Units_Consumed_kWh using an if / else block.
2. Use *input.<field>* to read and assign form fields, *alert* to show a message and *cancel submit* to stop the save.

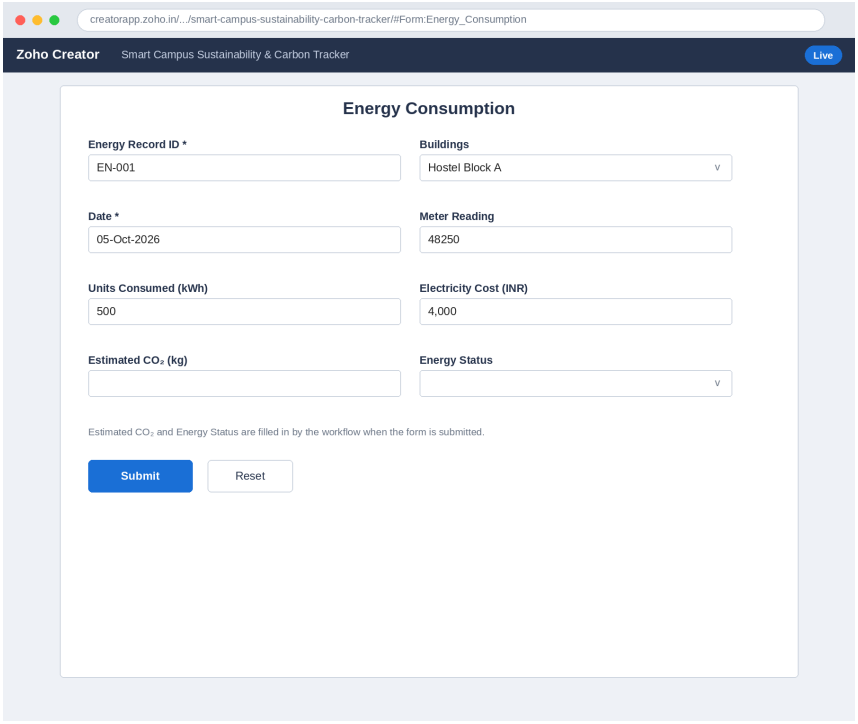
3. Check that braces are balanced and that every string is enclosed in quotation marks.

Step 5: Compiling and Saving

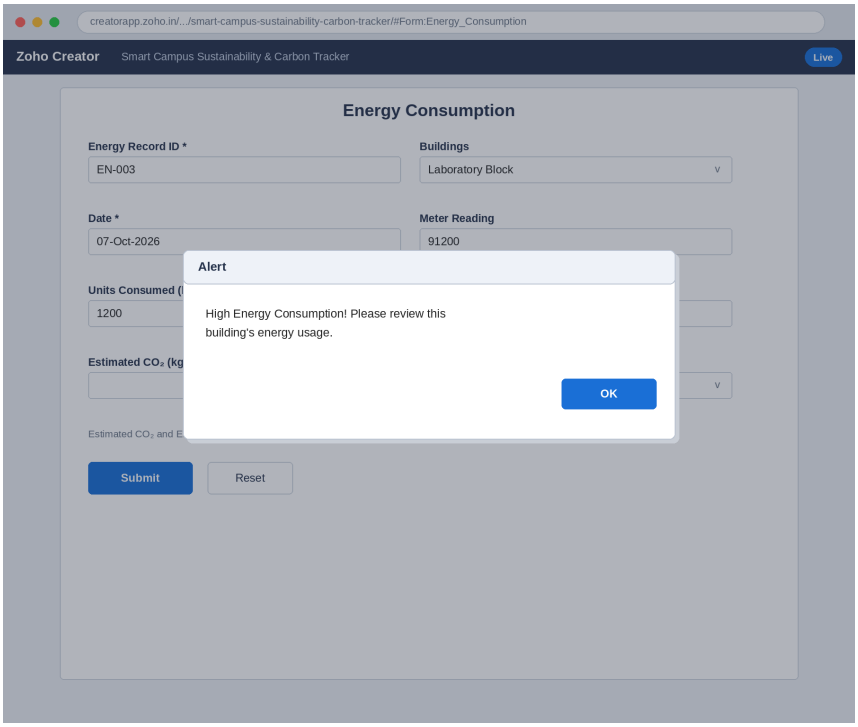
1. Review the code to eliminate syntax errors, mismatched brackets, or missing quotation marks.
2. Click Update to compile the code and push the workflow script live.

Step 6: Executing Live Application Testing

1. Switch the application from the editor to the live (Access) mode.
2. Open the live Energy Consumption form.
3. Enter Sample Record 1 (Hostel Block A, 500 kWh, cost 4,000) and click Submit.
4. Enter Sample Record 3 with 1200 kWh to test the alert, then correct the value and save.



Live Energy_Consumption form with Sample Record 1 entered (illustration).

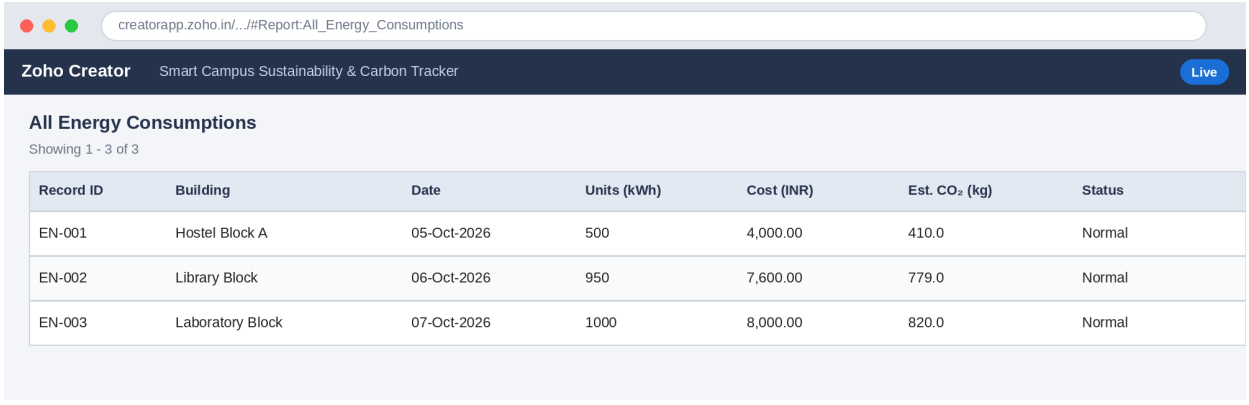


Alert displayed for 1200 kWh; the submission is cancelled and nothing is saved (illustration).

Step 7: Validation and Debugging Analysis

1. Verify that the record was added to the All Energy Consumptions report.
2. Check that Estimated CO_2 equals units × 0.82 and that Energy Status is correct.
3. Confirm that a reading above 1000 kWh is rejected with the alert message.

3. Output



All Energy Consumptions report after testing (illustration).

Test	Input (kWh)	Expected result	Observed result
T1: normal reading	500	Estimated CO_2 = 410.0; Status = Normal	Record saved with 410.0 and Normal
T2: high but allowed	950	Estimated CO_2 = 779.0; Status = Normal	Record saved with 779.0 and Normal
T3: above limit	1200	Alert shown; submission cancelled	Alert shown; no record created

Test	Input (kWh)	Expected result	Observed result
T4: boundary	1000	Estimated CO_2 = 820.0; Status = Normal (limit is “more than 1000”)	Record saved with 820.0 and Normal
T5: no units entered	blank / 0	Estimated CO_2 = 0; Status = Normal	Record saved with 0 and Normal

Debugging Analysis

Possible problem	Cause	Fix applied
Estimated CO_2 stays 0	Display name used instead of the link name of the field	Use the link name Units_Consumed_kWh in the script
Status not updated on edit	Workflow set to Created only	Set the Record Event to Created or Edited
Alert shown but record still saved	cancel submit missing	Add cancel submit after the alert
Script will not save	Missing brace or quotation mark	Re-check brackets and strings, then click Update

Result

The workflows were activated on the Energy_Consumption form without compilation errors, and the Deluge code passed the syntax checks. Filling out and submitting the live form generated a new record with an automatically calculated Estimated CO_2 and Energy Status, and readings above 1000 kWh were stopped by the alert. By combining form fields with targeted Deluge workflows, the Smart Campus Sustainability & Carbon Tracker moves from a passive data-collection form to an active system that validates and enriches every record at the moment it is entered.